

Gowtham Ayna Raveendran

Portland, OR, USA

☎ +1 (518) 763 6780 • ✉ argowtham@live.com • [in argowtham](#) • [rgowtham](#)

Summary

Data Scientist with 10+ years building production ML systems at scale - specializing in Anomaly Detection, NLP and MLOps. Patented work in Churn Detection; deployed models processing 100M+ records/day. Proficient in Python, SQL, Spark and AWS with expertise in model interpretability, observability and end-to-end MLOps pipelines.

Professional Experience

Yahoo

Data Scientist II - Traffic Protection

Sep'2021 – Present

- Architected and Implemented a distributed unsupervised machine learning pipeline using **PySpark (PCA, HDBSCAN)** to automatically discover and catalog complex botnet behaviors across millions of web requests, achieving 95+% variance capture and 100% cluster homogeneity when scaling from a local MVP to a Hadoop Grid.
- Identified a critical upstream data flaw during validation of a CTV device level bid request throttling solution - preventing a misaligned production rollout and protecting an estimated \$250k/year in operating costs without impacting ad spend
- Designed **model interpretability and observability pipeline** for Real-time scoring model to surface prediction drivers and to detect model drift, establishing proactive model health monitoring and reducing root cause analysis time by **75%**
- Leading cross-functional working group on data profiling IP intelligence dataset, refining VPN, Residential Proxy and CGNAT IP Classification to reduce both fraudulent traffic and false positives across advertising, search and edge platform.

Verizon Media

Data Scientist

Jan '18 – Sep'21

- Implemented a **CNN-LSTM-CRF Deep learning model** and PySpark pipeline for automated retraining and deployment of **Named Entity Recognition** system, processing over **100M+ records/day**
- Developed patented **Brand Churn Detection model** (RFM Segmentation) generating **\$1M** in revenue influence through targeted ad segments - US Patent US20230043820A1, Granted Feb 2023.
- Built a hierarchical **product category classifier** using multi-level logistic regression to map items to Google's product taxonomy; leveraged NER-derived features (brand, product line) and optimized training with **Vowpal Wabbit**, improving scalability.
- Developed and published a user-brand affinity framework by constructing proxy labels from behavioral signals, generating affinity scores to capture consumer preference beyond purchase data - Defensive Publication - IPCOM000277865D.

TrueScripts Management Services

Data Analyst

Jun '17 – Jan '18

- Built reporting dashboards in **R Shiny** and **Tableau** reducing turn around time by **75%**
- Engineered and Automated drug repricing system using **R** for estimating potential savings for new clients
- Architected and Built the company's centralized **data warehouse**, standardizing data access across departments.

Patents and Publications

- **Method and system for user group determination, churn identification and content selection:** US20230043820A1
- **Method and System for User Brand Affinity Modelling:** IPCOM000277865D

Education

- **M.S. in Data Science:** *Indiana University Bloomington* - GPA: 3.84/4.0 - May '17
- **B.E. in Electronics & Communication:** *Anna University, India* - GPA: 8.36/10 - Jun '12

Skills

Machine Learning: Keras, Scikit-learn, Decision Trees, PCA, Ensemble learning - Random Forests, XGBoost, Autoencoders, One class SVMs, Isolation Forests, DBSCAN, K-Means, mlflow, glum

Tools & Technologies: Airflow, Oozie, MLflow, Docker, Git, Looker, Otel, Grafana Tableau, Terraform, Spark, Hadoop, Hive, Vertica, Presto, Athena, Amazon Bedrock Agentcore, SageMaker, EMR, Athena, BigQuery(GCP)

Languages: Python, SQL, R, Bash, Java, Pig, Scala

Visualization & Monitoring: Tableau, Looker, Grafana, OpenTelemetry (OTEL)